

In-Class Exercise 1 · Solutions

Full house counting (instructor key — not for students)

Riddle meaning. A full house: three cards of one rank and two cards of a different rank.

Count.

$$\begin{aligned} N &= C(13, 1) C(4, 3) C(12, 1) C(4, 2) \\ &= 13 \times 4 \times 12 \times 6 \\ &= 3744. \end{aligned}$$

- $C(13, 1)$: choose the rank for the three-of-a-kind.
- $C(4, 3)$: choose 3 suits out of 4 for that rank.
- $C(12, 1)$: choose a different rank for the pair.
- $C(4, 2)$: choose 2 suits out of 4 for the pair.

Note. Do not multiply by an extra factor for “which block is the triple / which is the pair”: the two ranks are already chosen in order (triple rank first, pair rank second), so each full house is counted once.

Optional probability.

$$C(52, 5) = 2\,598\,960, \quad P(\text{full house}) = \frac{3744}{C(52, 5)} = \frac{3744}{2\,598\,960} = \frac{6}{4165}.$$

(The simplified fraction is optional for students.)

Quick check

Final answer for Task 2: **3744** full-house hands.